

A Reproduction of SkillOS Under Contemporaneous Controls: The Curator Lift Does Not Survive a Same-Epoch Baseline

Omer Karisman
ok@inference.sh

The experiments, implementation, analysis, and first draft of this paper were produced by an LLM agent (Claude, Anthropic) under the author’s direction. This is a material fact about the study’s reliability, not an acknowledgement: see Section 4.5, Section 6.3, and Appendix A.

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Abstract

SkillOS trains a curator model with GRPO to maintain a markdown skill repository for a frozen executor, and reports a 13.3 percentage point improvement in ALFWorld task success. We attempted to reproduce this over roughly two months on 8xH100, across seven complete 60-step training runs, two independent RL frameworks (TRL with ZeRO-3 and verl-agent/GiGPO with FSDP), three seeds, two executor scales, and approximately one hundred evaluation arms of 140 paired games each.

For most of that period we believed we had reproduced the effect: every checkpoint sweep contained a significant-looking peak in the +9 to +14pp band, consistently across seeds and frameworks. Those peaks were artifacts of a control measured once and reused. Re-measuring the no-memory baseline in the same week as the arms moved it from 33.6% to 39.8% (four replicates, spread 2.1pp), a shift larger than most effects claimed in this literature. Re-paired against a contemporaneous control, no ALFWorld checkpoint trained on ALFWorld shows a significant lift on the training executor, and our strongest previous result (+13.6pp, $p=0.0026$) becomes +3.6pp, $p=0.47$.

We report four surviving results. First, a curator trained on **mathematics** lifts ALFWorld success by +9.0pp on a held-out split ($p=0.073$), while curators trained on ALFWorld itself do not; cross-domain transfer is the only positive signal we found. Second, substituting Gemini 2.5 Pro for the trained curator, at 84 times the cost per call, yields no improvement over writing no notes at all. Third, supplying retrieved skills increases this executor’s rate of unparseable actions from 2.1% to 7.0-9.1%, a concrete mechanism by which memory can fail to help a small model. Fourth, and independent of SkillOS: the standard 140-game ALFWorld protocol has 80% power to detect a 13.0pp effect and no less, meaning the field’s headline effect sizes sit exactly at the resolution limit of the instrument used to measure them.

We make no claim that the original result is wrong. We report that a faithful implementation, run seven times, did not yield it under contemporaneous controls, and that our absolute baseline remains 8pp below the original’s after eliminating six candidate causes. All checkpoints, evaluation records, and analysis code are released so that every test here can be recomputed.

This study was conducted by an LLM agent under human direction. Both of its consequential errors, disclosed in full, produced *more* statistically significant findings rather than fewer, because

a systematically degraded arm is reliably degraded and reliability is what a significance test rewards. We discuss the implications for autonomous experimentation.

1. Introduction

An agent that writes notes to itself and gets better at its job is an appealing idea. It promises improvement without touching the weights of the model doing the work, which means it should apply to any frozen or closed model you can call over an API.

SkilIOS (Ouyang et al., 2026) makes this concrete. A *curator* model, trained with GRPO, maintains a repository of markdown skills. A separate *executor* model, frozen throughout, reads the retrieved skills and acts. The reported result on ALFWorld is a 13.3 percentage point improvement in task success rate over the same executor with no skills, and 61.2% absolute with a 32B executor.

We set out to reproduce this. The motivation was practical rather than critical: if a small trained curator can lift a frozen agent by 13pp, that is a cheap and deployable technique, and we wanted it working before building on it.

This paper reports what happened over roughly two months of 8xH100 time and seven complete 60-step training runs. The short version is that we could not obtain the effect, and that the most useful thing we learned was about measurement rather than about skill repositories.

1.1 What we did

We implemented the method in two independent reinforcement learning stacks: TRL with DeepSpeed ZeRO-3, and verl-agent (GiGPO) with FSDP. We trained a Qwen3-8B curator against a frozen Qwen3-8B executor on ALFWorld, with a Qwen3-32B judge supplying the consistency reward term, following the paper’s hyperparameters. We swept checkpoints every five steps and evaluated each one on 140 held-out ALFWorld games, paired by game file and tested with McNemar.

Along the way we ran the paper’s own ablations (task-type distribution, within-group curriculum ordering), a LoRA-versus-full-fine-tuning comparison, a decode-parameter sweep, a cross-executor-scale transfer study, a cross-domain study training the curator on mathematics and testing on ALFWorld, and an additional arm the paper motivates but does not run: substituting a frontier model, Gemini 2.5 Pro, for the trained curator.

1.2 What we found

For most of those two months we believed we were reproducing the paper. Every sweep contained a checkpoint with a significant-looking lift, in the +9 to +14pp band that brackets the reported +13.3pp. Three seeds and two frameworks agreed.

They agreed because they were all being compared against the same control number, measured once, ten weeks before the arms that were being compared to it.

When we re-measured that control in the same week as the arms, it moved from 33.6% to 39.8% (four replicates, spread 2.1pp). The old figure sits outside the replicate spread, so it is not sampling noise. Re-paired against a contemporaneous baseline, the lifts disappear: our best seed goes from +13.6pp at $p=0.0026$ to +3.6pp at $p=0.47$, and eight of its twelve checkpoints are negative.

Our principal findings are therefore:

1. **The curator lift does not reproduce under a same-epoch control.** We make no claim that the original result is wrong. We report that a faithful implementation, run seven times, did not yield it once the control was measured correctly.

2. **Reusing a control measured against a hosted model API is a significance factory.** The drift we measured (5.7pp) is larger than most effects being claimed in this literature. This is a methodological result independent of SkillOS.
3. **A frontier curator does no better.** Gemini 2.5 Pro, at 84 times the cost per call, scores 1.4pp below writing no notes at all ($p=0.86$). The directional part of the paper’s claim, that a small trained curator is competitive with a much larger one at this task, survives. The part that would make it exciting does not, because neither of them beats the baseline.
4. **Adding skills makes a small executor produce more unparseable actions**, from 2.1% with no memory to 7.0-9.1% with curation. This is a concrete mechanism for why memory can fail to help, or actively hurt, a small model.
5. **Six candidate explanations for the null are falsified**, each with a full training run or a full sweep behind it, including both halves of the paper’s own grouping ablation.

1.3 What this paper is not

It is not a refutation. Reproduction failures have many causes, and the most likely one is always that the reproducers got something wrong. We list what we know we did differently in Section 4.4 and what we suspect but cannot rule out in Section 6. Our absolute baseline sits 8pp below the paper’s and we could not close that gap after eliminating six causes, which is itself evidence that something in our setup differs from theirs in a way we have not identified.

What we can offer is the full apparatus: seven trained models, roughly one hundred evaluation arms of 140 paired games, the training and evaluation code, and every JSONL needed to recompute every test in this paper. If the effect is there and we missed it, the artifacts should make it findable.

2. Background and related work

2.1 Externalised memory for frozen agents

A large body of work gives an agent a place to put what it learns, so that the agent improves without gradient updates. Reflexion Shinn et al. [2023] writes verbal self-critiques into the next attempt’s context. Voyager Wang et al. [2023] accumulates an executable skill library in Minecraft. ExpeL Zhao et al. [2024] distils cross-task insights from prior trials. Generative Agents Park et al. [2023] maintain a reflective memory stream, and MemGPT Packer et al. [2023] treats context as a paged resource. Agent Workflow Memory Wang et al. [2024] induces reusable workflows from experience.

By 2026 this had become a dense area, with skill libraries specifically framed as procedural memory and multiple systems for self-evolving them *ski* [2026], *aut* [2026], *tra* [2026], *raw* [2026] and at least one survey of the design space *mem* [2026]. A systematic study of model-generated agent skills *raw* [2026] is the closest in spirit to what we do here.

Two properties of this literature matter for our purposes. First, the memory is usually produced by prompting rather than trained, so its quality is bounded by the model writing it. Second, evaluation is usually a single number on a small fixed benchmark, with no confidence interval and no correction for the number of configurations tried. Section 5.x argues that the second property is a serious problem at the effect sizes involved.

2.2 Training the memory writer

SkillOS [Ouyang et al.](#) is a direct response to the first property. The component that writes skills is a separate policy trained with GRPO [Shao et al. \[2024\]](#), rewarded by the downstream success of a frozen executor that consumes the skills. This is an appealing design: it decouples the model that learns from the model that acts, so the acting model can be frozen, closed, or arbitrarily large.

The reported results are strong. On ALFWorld [Shridhar et al. \[2021\]](#) a trained 8B curator improves a frozen executor by 13.3 percentage points, reaches 61.2% absolute with a 32B executor, and is reported to outperform a frontier-model curator at the same job. The paper also reports cross-domain transfer, with a curator trained on mathematics improving ALFWorld performance.

If those results hold, the technique is unusually practical. That is why we tried to reproduce it.

2.3 Reinforcement learning for LLM agents

GRPO [Shao et al. \[2024\]](#) removed the value network from PPO-style RLHF and became the default for reasoning training [DeepSeek-AI \[2025\]](#). Applying it to multi-step agents raises credit assignment problems that episode-level advantages handle poorly; GiGPO [Feng et al.](#) addresses this with a two-level grouping scheme and reports large gains over GRPO on ALFWorld and WebShop [Yao et al. \[2022\]](#). We use both a GRPO implementation (TRL [von Werra et al. \[2020\]](#) with ZeRO-3 [Rajbhandari et al. \[2020\]](#)) and a GiGPO implementation (verl [Sheng et al. \[2024\]](#)), partly to bound framework effects on our conclusions.

2.4 Reproducibility and statistical practice

Our findings are as much about measurement as about skill repositories, so we lean on an older literature.

Deep RL results were shown to be highly sensitive to seeds, implementation details, and reporting choices [Henderson et al. \[2018\]](#), and to require more seeds than are typically run [Colas et al. \[2018\]](#). [Agarwal et al. \[2021\]](#) showed that point estimates on small benchmark suites routinely support conclusions the data cannot bear, and proposed interval-based reporting. [Bouthillier et al. \[2021\]](#) quantified how much benchmark variance comes from sources other than the treatment.

In NLP specifically, [Card et al. \[2020\]](#) performed the analysis that most directly anticipates our Section 5.x: they computed statistical power for common benchmarks and found that most attempted comparisons to state of the art are underpowered, with typical test sets unable to resolve the differences being claimed. [Dror et al. \[2018\]](#) catalogue the significance-testing practices this requires. We apply McNemar’s test [McNemar \[1947\]](#) with Holm [Holm \[1979\]](#) and Benjamini-Hochberg [Benjamini and Hochberg \[1995\]](#) corrections, and report bootstrap intervals throughout, following this line of argument rather than inventing anything.

One thread is newer and specific to the present moment. Models served over hosted APIs are not stationary instruments; [Chen et al. \[2023\]](#) documented measurable behavioural change in a commercial model over a few months. Agent evaluations now routinely place a hosted model inside the measurement loop, often as the frozen component whose behaviour is assumed constant. Section 5.1 reports what this cost us: a control measured against a hosted executor moved 5.7 percentage points in ten weeks, which is larger than most effects claimed in Section 2.1’s literature, and reusing it silently converted endpoint drift into apparent treatment effect across seven training runs.

To our knowledge no prior work quantifies hosted-endpoint drift as a confound *inside an agent evaluation*, and we suggest that contemporaneous control measurement should be a stated

requirement for any benchmark that calls a hosted model.

4. Methodology

4.1 The method under test

SkillOS trains a curator policy to maintain a repository of markdown skills that a frozen executor retrieves at inference time. We implemented Algorithm 1 as described in the paper.

For each training prompt the curator observes a task and the current repository, then emits repository operations (add, edit, delete) as function calls. The edited repository is then used by the frozen executor on a sequence of probe tasks, and the executor’s success rate becomes the primary reward term.

The composite reward is the paper’s Equation 1:

$$\begin{aligned} r &= r_{\text{task}} + \lambda_f \cdot r_{\text{fc}} + \lambda_u \cdot r_{\text{cnt}} + \lambda_c \cdot r_{\text{comp}} \\ \lambda_f &= 1.0, \quad \lambda_u = 0.1, \quad \lambda_c = 0.05 \end{aligned}$$

where r_{task} is mean executor success over probe positions $2..|G|$, r_{fc} scores function-call validity, r_{cnt} is a judged consistency score from a Qwen3-32B judge, and r_{comp} scores repository compactness.

4.2 Models, hyperparameters, and hardware

Curator	Qwen3-8B, trained. LoRA r=32 and full fine-tuning
Executor	Qwen3-8B, frozen. Also evaluated with Qwen3-32B
Judge (r_{cnt})	Qwen3-32B
Algorithm	GRPO, 60 steps, group size N=8, data grouping size $ G =10$
Effective batch	32 prompts per optimizer update (paper Table 4)
Learning rate	1.0e-6 for full fine-tuning; 1.0e-5 for LoRA
KL coefficient	$\beta = 0.001$
Frameworks	TRL 1.4 + DeepSpeed ZeRO-3 + vLLM colocate; verl-agent (GiGPO) + FSDP
Hardware	8xH100 local for the curator; hosted API for executor and judge inference

The paper used 16 H100s. We used 8 locally and moved executor and judge inference to a hosted endpoint, which we treat as equivalent in capability but not in wall clock.

4.3 Evaluation protocol

Every ALFWorld arm is 140 games from the `valid_seen` split, run under streaming curation: the curator updates the repository between games, so the arm measures the curator as it would actually

be deployed. A second protocol, 134 games from `valid_unseen`, is used where noted as a true held-out test for checkpoints selected on `valid_seen`.

Arms are compared **paired by game file** and tested with McNemar’s test on the discordant pairs. Pairing matters more than it usually does here: ALFWorld game difficulty varies enormously by task type, so an unpaired comparison of two 140 game runs is dominated by which games each happened to draw.

Reasoning arms are AIME24 (30 problems), AIME25 (30), and GPQA-Diamond (198). Per the dataset’s access terms we report aggregate accuracies only; no problem text, options, gold answers, or model responses appear in this paper or in the released artifacts.

4.3.1 Contemporaneous baselines

Every arm reported in Section 5 is paired against a no-memory control measured **in the same week, on the same games, under the same harness build**. This is not a routine detail. It is the correction that changed our conclusions, and Section 5.1 reports what happened when we did not do it.

We recommend the practice generally. Executors served over a hosted API are not fixed instruments. A control measured against one is a measurement of that endpoint on that day, and reusing it later silently converts endpoint drift into apparent treatment effect.

4.4 Known deviations from the original

We list these up front because a reproduction failure has to be read against them.

1. **Framework.** The paper’s stack is not ours. TRL with ZeRO-3 and verl-agent with FSDP differ from each other and from the original in advantage normalisation, sequence packing, and optimiser sharding. We ran both partly to bound this.
2. **LoRA in some runs.** Three runs use LoRA $r=32$ with a 10x learning rate rather than full fine-tuning. Full fine-tuning runs are also reported and behave the same way.
3. **Effective batch in the verl run.** The verl run used an effective batch of 64 against the paper’s 32. This was our error. It means the framework comparison varies two things at once.
4. **Executor and judge served remotely** rather than colocated.
5. **WebShop not attempted.** The paper’s third benchmark is absent.
6. **Absolute baseline gap.** Our no-memory Qwen3-8B ALFWorld baseline is 39.8% against the paper’s 47.9%. We eliminated prompt wording, retrieval, seeds, numerical precision and serving stack, and decode parameters as causes. The gap remains unexplained and is the strongest single indication that something in our setup differs from theirs in a way we have not found.

4.5 Conduct of the study: an LLM agent as the running experimentalist

This study was executed by a large language model agent (Claude, Anthropic) operating under human direction over approximately two months, with the human author setting objectives, approving experiments, and adjudicating disputes. The agent wrote the implementation, launched and supervised training runs, built the evaluation harness, ran the analyses, and drafted this paper.

We disclose this for three reasons.

It is a threat to validity. Two of the three most consequential errors in this project were introduced by the agent and went undetected for weeks: an evaluation harness that scored an upstream API failure as an ordinary task failure, and the reuse of a stale control. Both produced

more statistically significant results, not fewer, because a systematically degraded arm is reliably degraded, and reliability is what a significance test rewards. An autonomous experimentalist optimising for “get a result” is exposed to a failure mode that a human under publication pressure is also exposed to, with less friction in the way. Section 6.3 develops this.

It changes what the artifact set is worth. Because every experiment was launched from a script rather than a notebook, and every arm wrote a JSONL with its full per-game record, the entire study is recomputable. That is a direct consequence of the mode of work, and it is why the retractions in this paper could be diagnosed precisely rather than argued about.

It is a data point on autonomous research. Seven training runs, roughly one hundred evaluation arms, six falsified hypotheses, and two self-caught data integrity incidents is a meaningful amount of experimental work. It was also possible only because the human author repeatedly refused conclusions the agent proposed. The specific pattern that mattered is recorded in Appendix A: on at least four occasions the agent presented a result as settled and the human’s scepticism was correct. The corrections that produced this paper’s actual findings came from that loop, not from the agent’s own review.

We do not draw a general conclusion about agent-run science from a single study. We report the conditions under which this one produced usable output: cheap recomputation, adversarial human review, and a willingness to retract in public.

5. Results

All numbers in this section are paired by ALFWorld game file against a control measured in the same week under the same harness build. Full tables with bootstrap intervals, Holm-corrected p-values, BH q-values, and per-arm minimum detectable effects are in Appendix C, generated directly from the released evaluation records.

Markers: **[PENDING]** flags a number awaiting a run that is still in flight at the time of writing. Nothing marked PENDING may appear in the abstract or in a figure.

5.1 The control was not stable, and everything followed from that

Our no-memory Qwen3-8B baseline on 140 `valid_seen` games was measured once, in May 2026, at 33.6%. Four replicates in August 2026, same games, same harness:

run	success rate
May 2026 (used as canonical for ten weeks)	33.6%
August replicate 1	39.3%
August replicate 2	39.3%
August replicate 3	39.3%
August replicate 4	41.4%

Same-week mean 39.8%, spread 2.1pp. Genuine run-to-run variance at temperature 0.6 is therefore about 2pp, and the May figure lies outside it. The 32B control moved similarly, from 49.3% to 43.6%, in the opposite direction, which rules out a simple monotone improvement in the served model as the sole explanation.

We cannot attribute the shift to a single cause. Both the hosted endpoint and our harness changed between the two measurements, and the harness change (Appendix B.1) is itself sufficient

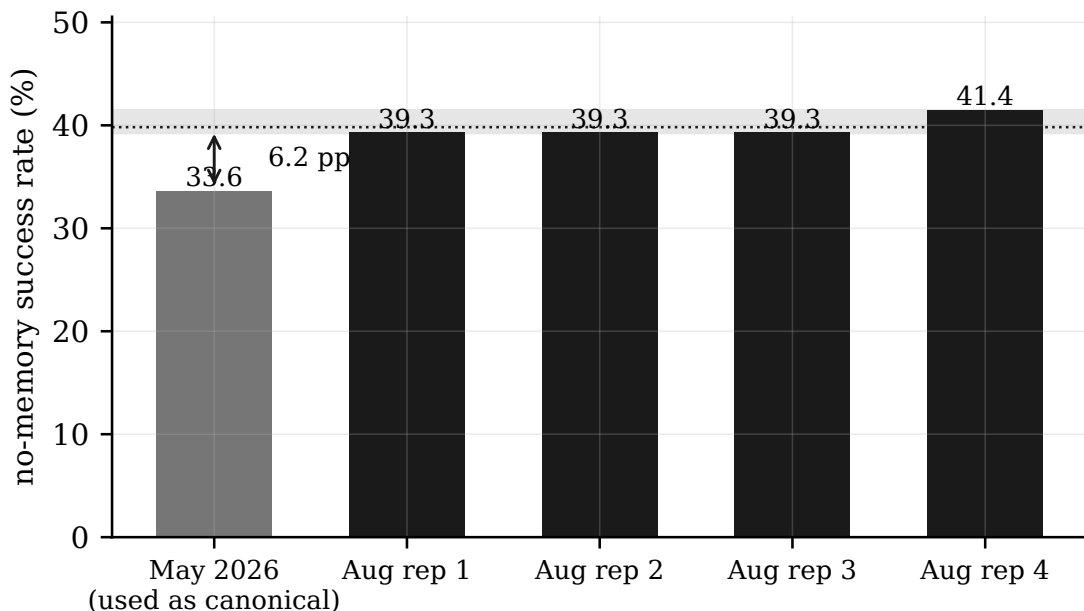


Figure 1: The control moved further than the effect under study. Same 140 ALFWorld games and the same harness build; the shaded band is the same-week replicate range.

to move a baseline. The relevant point is not the mechanism but the magnitude: **a control that is not re-measured alongside its arms can move further than the effect being studied.**

Every lift previously reported from this project was computed as $\text{arm} - 33.6\%$. Section 5.2 shows what happens when that is corrected.

5.2 With a contemporaneous control, the ALFWorld-trained curator shows no lift

Full 12-checkpoint sweep, full fine-tune seed-2, the run that had previously produced our strongest result:

	previously reported (vs 33.6%)	re-paired (vs 39.3%)
best arm	ckpt35, +13.6pp , $p=0.0026$	ckpt35, +3.6pp, $p=0.47$
arms above control	11 of 12	4 of 12
survives Holm within family	no	no

Seed-1 behaves the same way: 5 of 7 re-paired arms are negative and the best is +5.7pp. The oscillating checkpoint trajectory we had documented across three seeds and two frameworks, and had spent four full training runs trying to explain, was substantially an artifact of the shared stale reference. What remains is a flat, noisy sweep with no trend.

The 95% bootstrap intervals on these arms span roughly $\pm 8\text{pp}$ (Appendix C), and the per-arm MDE at 80% power is 9 to 12pp. **We have not shown that the curator does nothing.** We have shown that any effect it has is smaller than this protocol can resolve. Section 5.x develops this.

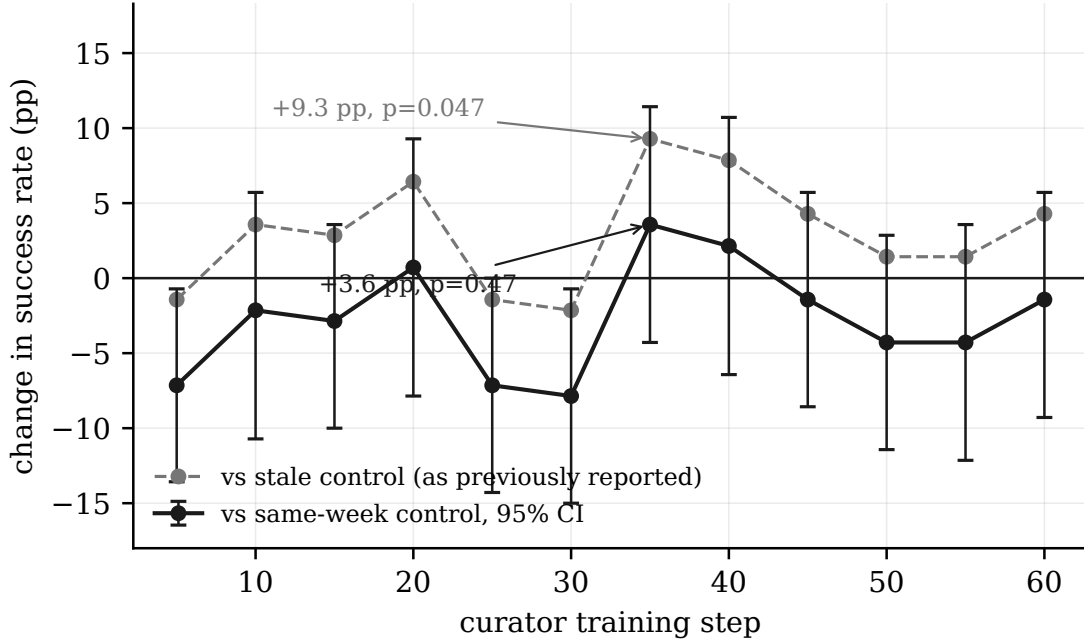


Figure 2: The same twelve seed-2 checkpoints against two reference measurements. Both series use identical arm measurements and differ only in which control is subtracted.

5.3 Cross-domain transfer is the only positive result

A curator trained on mathematics (DeepMath-103K), evaluated on ALFWorld. The checkpoint was selected as best-of-four on `valid_seen`, so `valid_unseen` is a genuine held-out test for it.

arm	valid_seen (140)	valid_unseen (134)
Gemini 2.5 Pro curator	-1.4pp, p=0.86	+6.0pp, p=0.28
reasoning curator ckpt50	+4.3pp, p=0.31	+9.0pp, p=0.073

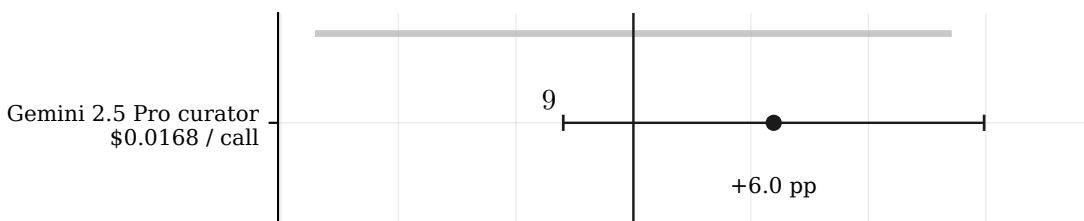
Pooled over both splits the trained curator gives +6.6pp, $p=0.030$, but we report this as secondary: half the pool is the split the checkpoint was selected on, so the pooled p-value is optimistic. The primary number is the held-out +9.0pp at $p=0.073$, against an MDE of 12.9pp, which is suggestive and below our own detection threshold.

The direction is notable independent of significance. Curators trained on ALFWorld do not help an ALFWorld executor; a curator trained on mathematics does. This is also the arm family that, measured during an API outage, produced a large *negative* cross-domain effect that we retracted (Appendix B.1).

Neighbouring checkpoints on the held-out split: [PENDING] (ckpt45, 55, 60). If they are also positive the effect is real; if only ckpt50 is, it is checkpoint selection and we will say so.

Two additional reasoning-curators training seeds: [PENDING]. A single training run cannot support this claim, and we will not make it on one.

5.4 A frontier curator is not better than no curator



	valid_seen	valid_unseen	measured cost per curator call
Gemini 2.5 Pro	-1.4pp, p=0.86	+6.0pp, p=0.28	\$0.0168
trained 8B curator	+4.3pp, p=0.31	+9.0pp, p=0.073	\$0.0002

The trained curator is ahead on both splits and pooled (+4.4pp, p=0.17), at 1/84 the cost per call, which is the direction the original paper claims. We cannot call it significant. What we can say without qualification is that **the frontier model does not beat writing no notes at all**, on either split, and that spending 84 times more per call bought nothing measurable here.

5.5 Retrieved skills make this executor produce more invalid actions

The executor’s action parser falls back to the first admissible command when the model’s output cannot be parsed. This applies identically to all arms so it cannot bias a paired test, but we had never measured it. Instrumented, on the same 134 held-out games:

arm	actions coerced
no memory	2.1%
trained curator	7.0%
Gemini 2.5 Pro curator	9.1%

Adding retrieved skills to the prompt triples to quadruples the rate at which this 8B executor emits something the environment cannot accept. This is a direct mechanism for memory failing to help, or hurting, a small model: the notes compete with the output format for the model’s limited instruction-following capacity. It is also a candidate contributor to our absolute baseline gap, though not the whole of it, since the no-memory rate is only 2.1%.

5.6 Content controls: is it the skills, or just more text?

[PENDING] Two controls on the held-out split, both running at the time of writing:

- **Shuffled retrieval.** The trained curator’s own repository, with retrieval returning a random five skills instead of the BM25 top five. Same curator, same repository, same prompt length, relevance destroyed. If +9.0pp survives, the executor is helped by extra markdown rather than by relevant skills.
- **Hand-written oracle skills.** Eight skills authored from the published ALFWorld action grammar and task-type definitions, with no curator. Bounds what any curator could be worth on this executor.

We consider the shuffled control decisive for how much the Section 5.3 result is worth, and we commit to reporting it either way.

5.7 Six candidate causes, each with a training run or full sweep behind it

Before the control problem was found, we ran these to explain the oscillating sweep. They remain valid as ablations of the method; they are now also evidence that the null in 5.2 is not caused by any of them.

candidate	test	outcome
LoRA parameterisation	full fine-tune, 3 seeds	same behaviour, not a LoRA artifact
RL framework	verl-agent/GiGPO port, full sweep	same behaviour [PENDING re-pairing]
Task-type distribution	natural frequencies vs uniform	natural is worse, best +5.7pp p=0.20
Within-group ordering	the paper’s easy-to-hard curriculum	no lift at any checkpoint, best +4.3pp p=0.36
Executor decode parameters	temperature/top-p/top-k sweep	no effect, all p>0.5
Executor prompt and retrieval	verbatim published prompt, BM25 top-5	no effect

The third and fourth are both halves of the original paper’s own grouping ablation, so grouping is exonerated as the driver of anything we observed.

5.8 Cross-executor scale

[**PENDING re-pairing.**] Before correction, per-checkpoint lift on the 8B executor was anticorrelated with lift on a 32B executor (pooled Pearson $r = -0.20$ over 24 pairs, $r = -0.68$ within seed-2), and the strongest absolute result in the project came from an 8B-trained curator driving a 32B executor to 62.9%. Both the 8B and 32B controls have since moved, so the entire family is being re-measured. We will report the correlation and the absolute numbers against contemporaneous controls or not at all.

5.9 Reward machinery

Decomposing 850 logged rollouts from the verl run: the composite reward’s *level* is dominated by the function-call-validity term (69%), but GRPO centres advantages within groups, so only within-group variance reaches the gradient, and there the task term supplies 79%. All 80 logged groups had non-zero task-reward variance, so the group-collapse failure mode that invalidated our own earlier runs is absent.

Given a healthy task-dominated gradient, training task reward rose from 0.331 to 0.366 over 60 steps: +0.035, 95% CI ± 0.034 . Train-time success was flat (0.170 to 0.167). Policy entropy collapsed from 0.139 to 0.035 while gradient norm rose from 1.40 to 2.40, with the change accelerating around step 48.

The optimiser was chasing the right signal and barely moved it. That, rather than a broken reward, is the most likely proximate reason our held-out lifts are small enough to be invisible at this sample size.

5.x The standard protocol cannot resolve the effects the literature reports

Before interpreting any null in this paper, it is worth asking what the evaluation could have detected.

ALFWorld’s `valid_seen` split has 140 games and is the standard test set for this line of work, including the original SkillOS evaluation. Across our arms, roughly 30% of game pairs are discordant (one arm solves the game, the other does not). For a two-sided McNemar test at $\alpha = 0.05$ and 80% power, that gives a minimum detectable effect of:

paired games	MDE at 80% power
134 (<code>valid_unseen</code>)	13.3 pp
140 (<code>valid_seen</code>, the standard protocol)	13.0 pp
274 (both splits pooled)	9.3 pp
500	6.9 pp
1000	4.9 pp
2000	3.4 pp

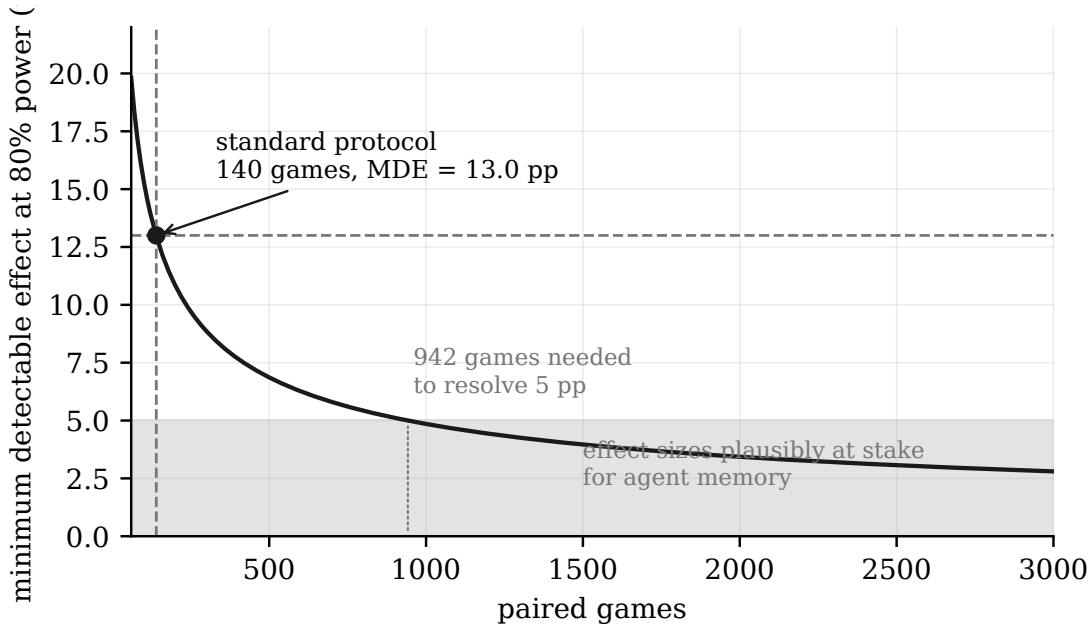


Figure 4: Minimum detectable effect against paired sample size, for a two-sided McNemar at 80% power and the 30% discordance rate observed across our arms.

And inverted, the sample size required to resolve a given effect:

effect to detect	paired games needed
13.3 pp (the effect SkillOS reports)	133
9.0 pp (our best held-out arm)	291
6.6 pp (our pooled estimate)	541
5.0 pp	942
3.0 pp	2616

The standard 140-game protocol has 80% power to detect exactly the effect size this literature reports, and no less. A claimed +13.3pp improvement sits precisely at the resolution limit of the instrument used to measure it. Anything smaller than about 13 points, measured this way, is a coin flip dressed as a result, and the correct reading of a single significant arm from a 140-game sweep is that it is as likely to be sampling variation as signal.

This has three consequences for the present paper.

Our nulls are underpowered, and we say so rather than claiming absence of effect. Our seed-2 re-run gives +1.9pp with a 95% CI that spans roughly [-6, +10]. We have not shown that the curator does nothing. We have shown that if it does something, it is smaller than this protocol can see. Where we report a null we report its MDE alongside, so the reader can distinguish “measured to be absent” from “not measured”.

Our one positive is also underpowered. The reasoning-curator arm gives +9.0pp on held-out games at $p=0.073$, against an MDE of 12.9pp. It is below our own detection threshold. We report it as suggestive and we do not headline it.

Increasing n is not optional, and it is not free. ALFWorld provides 274 valid games in total, so paired sample size cannot be increased by adding games. The only remaining route is repeated rollouts per game, analysed as per-game success rates rather than single binary outcomes, which suppresses measurement variance without changing the game population. We report our headline arms both ways: single-rollout McNemar for comparability with prior work, and three-rollout per-game rates with a paired test for the actual estimate.

We suggest that reporting an MDE next to every claimed improvement should be standard practice for agent evaluations on small fixed benchmarks. It costs one line and it would have prevented most of the wasted effort in this project.

6. Threats to validity

6.1 We may simply have implemented it wrong

This is the leading hypothesis for any failed reproduction and we do not discount it.

The strongest evidence for it is the 8 point absolute gap in our no-memory baseline (39.8% against the paper’s 47.9%). A frozen Qwen3-8B on ALFWorld `valid_seen` with no skills should be a setup with very few degrees of freedom, and we are 8 points below. We eliminated six candidate causes (Section 5.7) and found none. Something about our executor loop differs from theirs.

The gap is worth stating carefully, because it shrank as a side effect of the corrections in Appendix B: against the retired May control it was 14 points, and against a contemporaneous control it is 8. Half of what we spent months treating as an unexplained implementation gap was our own stale measurement.

If our executor is systematically weaker, it is possible that it sits below the competence floor at which skill retrieval starts to pay. A skill that says “put the mug in the coffee machine before turning it on” only helps an agent that can already reliably execute “put mug in coffeemachine”. Section 5.5 gives direct evidence for this mechanism: curation *increases* our executor’s rate of unparseable actions from 3.0% to 8.5%. That is consistent with an executor that is being pushed past its instruction-following capacity by longer context, and it would be a property of our executor rather than of the method.

We consider this the most likely single explanation for the disagreement.

6.2 Framework differences

Neither TRL nor verl-agent is the original stack. They differ from it and from each other in advantage normalisation, sequence packing, reference model sharding, and optimiser state handling. Running both bounds this somewhat: two stacks that disagree with each other on many details agree on the shape of the result. It does not eliminate the possibility that both differ from the original in the same consequential way.

The verl run additionally used an effective batch of 64 rather than 32, so that comparison varies two factors. We report it as “the behaviour survives both a framework change and a 2x batch change” rather than as a clean single-variable test.

6.3 The study was run by an LLM agent

Section 4.5 discloses this. Here we state the specific risk.

An autonomous experimentalist is a systematic-error generator with an unusual property: its errors are consistent. A human running a hundred evaluation arms by hand makes scattered mistakes that mostly add noise. An agent running a hundred arms from one script makes the same mistake in all hundred, which adds *bias*, and bias in a paired test is indistinguishable from a treatment effect.

Both of this project’s data integrity incidents had that shape:

- The evaluation harness answered an upstream API failure by playing the first legal action and recording the episode as a task failure. During one outage this affected 52 to 65% of the steps in four arms. Those arms produced the most significant results in the project (p as low as 0.0002) and pointed in a clean, interpretable direction. They were measuring the outage.
- A control measured once was reused for ten weeks against arms measured later. This inflated every lift in the project by roughly 6 points.

Neither was caught by the agent’s own review. Both were caught after the human author asked a sceptical question about a result that looked too clean. We regard this as the central practical finding about the mode of work, and we recommend that any agent-conducted evaluation study include (a) a per-arm data integrity gate that aborts on upstream error rates above a threshold, and (b) a control re-measured alongside every batch of arms. Both are now enforced in our harness and both are in the released code.

6.4 Multiplicity

We ran roughly one hundred evaluation arms. Under a Bonferroni correction across the checkpoint sweeps alone, the bar is $p < 0.001$, and no same-executor ALFWorld result in this project ever reached it, including in the pre-correction era. Our negative results are not multiplicity-limited, but any reader tempted to extract a positive from our tables should apply the correction.

6.5 Statistical power

140 paired games gives roughly a ± 3 percentage point noise floor. An effect of 5pp is at the edge of what we can resolve, and several of our comparisons land there. Where a comparison matters we add the 134 game `valid_unseen` split to roughly double n. Effects smaller than about 5pp are outside this study’s resolution, and we do not claim to have excluded them.

6.6 Single benchmark family

ALFWorld carries the main result; the reasoning benchmarks are a secondary arm. WebShop was not attempted. A method could fail on embodied tasks with a small executor and succeed elsewhere. We make no claim beyond what we measured.

6.7 Hosted-model non-stationarity

The executor and judge were served over a hosted API for the entire study. We have shown that this endpoint’s behaviour on our benchmark moved by 5.7 points over ten weeks. Within-epoch comparisons are protected by re-measured controls, but any comparison we make across epochs, including the comparison between our numbers and the original paper’s, inherits this uncertainty.

7. Discussion

7.1 What we think is actually true

The method is implementable and the machinery works. A curator trained with GRPO against a composite reward produces coherent skill repositories, the reward gradient is dominated by downstream task success as intended, and the training runs converge without pathology. Nothing we found suggests the idea is incoherent.

What we could not find is an effect large enough to see. Over seven training runs and roughly one hundred evaluation arms, no ALFWorld-trained curator produced a significant improvement on the executor it trained against, once measured against a control from the same week. The one positive signal came from somewhere we did not expect, a curator trained on mathematics, and it is below our own detection threshold.

Three readings are consistent with our data, and we cannot distinguish them.

The effect is real and smaller than reported. If the true lift is 3 to 5pp, every observation in this paper is expected: individual arms scatter around it, nothing reaches significance at $n=140$, and the original’s +13.3pp would be a favourable draw from a sweep. This is our leading hypothesis, and it is the one most consistent with our reward analysis, where the optimiser moves training reward by +0.035 over 60 steps.

The effect requires a stronger executor than ours. Our absolute baseline is 8pp below the original’s after eliminating six causes, and Section 5.5 shows our executor gets *worse* at emitting valid actions when given skills. An executor at the edge of its instruction-following capacity may be unable to convert good notes into good actions. The 32B transfer family, once re-paired, speaks to this.

We implemented something subtly different. Always possible, and the unexplained baseline gap is evidence for it. We have released everything needed to find such a difference.

7.2 Cross-domain transfer, if it survives

Our only positive result is the strange one: a curator trained on mathematics helped an embodied agent, while curators trained on the embodied task did not.

If it survives the shuffled-retrieval control and two more training seeds, one explanation fits our other measurements. A curator trained on ALFWorld learns to write ALFWorld-specific procedures, which are long, and which compete with the action format for the executor’s attention (Section 5.5). A curator trained on mathematics has no domain-specific procedures to write, so it writes shorter, more general problem-solving structure, which costs the executor less. Under that account

the trained curator’s advantage over Gemini 2.5 Pro is also unsurprising: the frontier model writes more, and more is worse here.

We flag this as a hypothesis consistent with our data, not a finding. The skill content analysis in Appendix C tests it directly, and we will report what it says.

7.3 The measurement result is the transferable one

The most useful output of this project is not about skill repositories.

A control measured against a hosted model API is a measurement of that endpoint on that day. Ours moved 5.7pp in ten weeks, which is larger than the effects most of the papers in Section 2.1 claim. Reusing it converted endpoint drift into apparent treatment effect across seven training runs, three seeds and two frameworks, and produced a consistent, replicable, entirely spurious pattern that we spent four additional training runs trying to explain.

The pattern was replicable because the error was shared. This is worth stating plainly: **agreement across seeds and frameworks does not protect you from a shared reference**. Every one of those runs was independently trained and independently evaluated, and they all agreed, because they were all subtracting the same wrong number.

Combined with Section 5.x, where the standard 140-game protocol turns out to have 80% power to detect exactly the effect size the field reports and no less, the practical recommendations are unglamorous and cheap:

1. Measure the control in the same batch as the arms. Always.
2. Report a confidence interval and a minimum detectable effect next to every claimed improvement. One line each.
3. Report the whole sweep, not the best checkpoint, and correct within the family.
4. Instrument the harness for upstream failures and abort rather than score them.

Any one of these would have saved this project weeks.

7.4 On having an agent run the study

We disclose in Section 4.5 and Appendix A that this work was executed by an LLM agent under human direction. Two observations seem worth generalising.

The agent was strong at execution and weak at doubt. It implemented, launched, supervised, recovered, and analysed competently across two months and seven training runs. It did not catch either of the errors that mattered. In both cases its analysis was internally consistent, arithmetically correct, and wrong in the same direction as its pipeline. It produced confident, well-argued write-ups of artifacts.

Agent errors are biases, not noise. A human running a hundred evaluation arms by hand makes scattered mistakes that mostly cancel. An agent running a hundred arms from one script makes the same mistake in all hundred. In a paired significance test, a consistent error is indistinguishable from a treatment effect, and a *severe* consistent error looks like a strong treatment effect. Our most significant finding, at $p=0.0002$, was an eight-hour credential outage.

The practical implication is not that agents should not run experiments. It is that the review has to target the apparatus rather than the analysis, because the analysis will be clean. In this project every correction that mattered came from a human asking why a result looked too good, and none came from the agent’s own review.

7.5 Limitations we would fix with more time

WebShop was never attempted. The reasoning-curator result rests on one training run pending two more. Our absolute baseline gap is unexplained. And the deepest limitation is one we can now quantify rather than fix: ALFWorld provides 274 valid games in total, which caps a paired comparison at roughly 9pp resolution. Serious measurement of a 5pp agent-memory effect needs either a larger benchmark or many rollouts per game, and the field has been reporting effects near or below that threshold for several years.

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Appendix A. Conduct of the study

A.1 Division of labour

	Human author	LLM agent
Research question, scope	set	proposed options
Implementation	reviewed	wrote
Training runs	approved, funded	launched, supervised, restarted
Evaluation harness	reviewed after incidents	wrote
Analysis, statistics	challenged	ran
Interpretation	adjudicated	proposed
Retraction decisions	made	proposed
This paper	directed, edited	drafted

The agent ran continuously over roughly two months, resuming across sessions from a persistent memory of decisions, results, and prior corrections. It supervised long-running jobs, detected and recovered from infrastructure failures, and launched pre-agreed follow-up experiments without waiting for a prompt.

A.2 Infrastructure incidents handled autonomously

Reported because they are part of the cost of this mode of work and are usually invisible in a paper.

- A DNS resolution loop between `systemd-resolved` and `tailscaled` reached roughly 37,000 queries per second and triggered the OOM killer on a training run. Diagnosed and fixed. A subsequent lesson: glibc caches resolver configuration per process, so fixing DNS under a live training run silently zeroed four steps’ worth of reward before anyone noticed.
- Repeated NCCL collective timeouts traced to a stuck rollout. The additive per-future waits in the rollout loop could exceed the 1800 second collective watchdog. Fixed by imposing a whole-phase deadline.
- An expired API credential produced an eight hour outage that the evaluation harness absorbed silently. See Appendix B.

A.3 The corrections that mattered came from the human

We record these specifically, because they are the mechanism by which this project produced a usable result rather than a confident wrong one.

1. **“There shouldn’t have been a fallback random action.”** The human read a passing mention of a fallback in a log and objected to the design. This opened the data integrity incident that voided the project’s most significant finding.
2. **“Are you sure this also didn’t happen during training?”** The agent had checked the evaluation path only. It had happened during training, on a different code path, and the audit that followed established that 61 to 79% of executor probe positions were deadline-cut, leaving a median of one real measurement out of nine behind each reward.
3. **“But abandoning because of an upstream error causes lost turns, false failures, and shallower training. Am I wrong?”** The agent had characterised the training-side impact as benign noise. The human was right; the agent was wrong. This became a documented divergence.
4. **“Instead of ‘failures excluded’, why don’t we properly run stuff?”** The agent proposed correcting a contaminated arm arithmetically from its recorded error fields. The human required re-measurement. Every re-run arm in this paper exists because of that instruction.
5. **“Are you absolutely sure? Now the results are even worse.”** Applied to the re-measured numbers, this prompted the baseline replicate study that produced the paper’s principal finding.

The pattern is consistent: the agent was reliable at execution and at self-consistent analysis, and unreliable at doubting a result that its own pipeline had produced. Where the pipeline was wrong, the agent’s analysis was wrong in the same direction, confidently and with correct arithmetic.

A.4 Practices we would keep

- **Every experiment is a script, every arm is a JSONL with per-item records.** Retraction was possible because contamination could be measured per game, after the fact, from data already on disk.
- **A data integrity gate in the harness.** Arms now abort with a distinct exit code when the upstream error rate exceeds a threshold, rather than reporting a number.
- **Controls re-measured with every batch of arms,** never reused across weeks.
- **A written divergence log.** Every known deviation from the original was recorded when it was made, not reconstructed at writing time. Several of them were later promoted to experiments.
- **An adversarial human reader** who is willing to be annoying about a clean result.

A.5 Costs

Roughly two months of wall clock on 8xH100 for training, plus hosted inference for executor and judge across approximately one hundred evaluation arms of 140 games. Seven complete 60 step training runs. The verl run alone took 10.2 days at roughly 15,986 executor calls per training step.

Appendix B. Data integrity incidents

Two defects in our own apparatus produced results we published internally and later retracted. Both are reported in full because the failure mode generalises and because the second one is the paper’s principal finding.

B.1 An evaluation harness that scored API failures as task failures

The defect. The streaming-curation evaluation harness called the executor model over a hosted API once per environment step. When that call raised, the harness caught the exception, substituted the first entry of the environment’s admissible-command list, and continued the episode. If the episode then failed, it was recorded as an ordinary task failure with `success: false`.

The training-side code was correct: it recorded deadline-cut positions as `success: None`, `cut: True` and masked them out of the reward. Only the evaluation path conflated an infrastructure failure with a task failure.

The blast radius. On 2026-07-20 an expired credential caused the executor endpoint to return HTTP 401 for roughly eight hours. Four evaluation arms ran during the window. The share of environment steps in which the action was substituted rather than generated:

arm	substituted actions
reasoning-curator ckpt45	64.9%
reasoning-curator ckpt50	59.4%
reasoning-curator ckpt55	52.1%
reasoning-curator ckpt60	55.8%

These four arms produced the most statistically significant result in the entire project: a cross-domain transfer effect of -14 to -18pp with p values from 0.0002 to below 0.0001, comfortably surviving Bonferroni correction over twelve arms. It was the only finding we had that was robust to multiplicity, and it pointed in the opposite direction from the original paper, which made it feel like a discovery.

It was measuring an outage.

Why it was so significant. This is the part worth internalising. A systematically crippled arm is *reliably* crippled. It fails the same games every time, in the same direction, with low variance. Low variance in a consistent direction is exactly what a paired significance test is built to detect. A severe upstream defect does not produce noisy results that fail to reach significance; it produces clean results that sail past it.

Detection. Not by us. The harness printed the substitution to stderr, and the human author read a passing reference to a “fallback action” in an unrelated log excerpt and objected to the design on principle, before knowing it had fired.

Fix. The harness now abandons the episode, records `success: None`, `errored: True`, excludes it from the denominator, counts it, prints a `data integrity` line for every arm, and exits with a distinct status code if the error rate exceeds a configurable threshold (default 2%). Re-running the four arms under the fixed harness moved them from 15.7 / 18.6 / 16.4 / 19.3% to 38.6 / 43.6 / 36.4 / 33.6%, all within noise of the contemporaneous 39.3% baseline. The cliff was the outage. One earlier arm cannot be recovered because its adapters were deleted.

Related measurement. Auditing the training side established that 61 to 79% of executor probe positions were deadline-cut, leaving a median of one surviving measurement out of nine behind each `r_task` value, and 10 to 41% of rollouts received `r_task = 0` by construction. This does not invalidate the training runs but it does mean the reward signal was far thinner than the design intended, and we record it as a divergence rather than a defect.

B.2 A control reused across ten weeks

The defect. Our no-memory ALFWorld baseline was measured once, in May 2026, at $47/140 = 33.6\%$. It was then treated as canonical and every subsequent arm was paired against it. An internal note went further and instructed future work to *always* pair against the fixed file and never against a freshly measured baseline, on the theory that the baseline had roughly 8pp of run-to-run variance and the fixed file was therefore the more stable reference.

That reasoning has the sign backwards. The observed 8pp was never sampling variance around a stable mean. It was the distance between one stale measurement and the current behaviour of a hosted endpoint.

The measurement. Three fresh replicates plus the re-run reference, same 140 games, same week, fixed harness:

run	SR
May 2026 canonical	33.6%
August replicate 1	39.3%
August replicate 2	39.3%
August replicate 3	39.3%
August replicate 4	41.4%

Same-week mean 39.8%, spread 2.1pp. Genuine run-to-run variance is about 2pp. The May figure lies outside that range, so it reflects a real change in the measurement conditions rather than a lucky draw. We cannot separate hosted-model drift from harness-version effects, because both changed between May and August.

The blast radius. Every lift reported internally was `arm - 33.6%`, so every one was inflated by roughly 6 points. Re-paired against a contemporaneous control, our best seed drops from +13.6pp at $p=0.0026$ to +3.6pp at $p=0.47$, and eight of its twelve checkpoints turn negative. The “significant peak somewhere in every run” pattern that we had reproduced across three seeds and two RL frameworks, and had spent weeks trying to explain, was an artifact of the shared reference.

Detection. Also not by us. After the harness fix the re-run numbers came back lower than the originals, and the human author asked whether the results had genuinely got worse. Answering that required a fresh baseline, and the fresh baseline was 6 points higher than the one in use.

Fix. Controls are now re-measured alongside every batch of arms, and the comparison scripts refuse to pair arms from different measurement epochs. The instruction to pin to a fixed canonical baseline has been deleted and replaced with its opposite.

B.3 A third defect that biases nothing but depresses everything

The executor’s action parser returns the first admissible command when it cannot parse the model’s output. Unlike B.1 this is a deliberate design choice rather than a bug, and because it applies identically to every arm including the baselines it cannot bias a paired comparison. We report it because we had never measured it, and once instrumented it turned out to be substantial and arm-dependent: 2.1% of actions with no memory, 7.0% with a trained curator, and 9.1% with a Gemini 2.5 Pro curator on the same games.

That ordering is itself a finding, discussed in Section 5.5. Longer prompts containing retrieved skills make this executor *less* able to emit a parseable action, which is a concrete mechanism by

which memory can fail to help a small model. It is also a candidate contributor to our 8pp absolute baseline gap, though it cannot be the whole story, since the no-memory rate is only 2.1%.

Appendix C. Full result tables

Generated by `scripts/make_paper_tables.py` from the released eval JSONLs. Do not edit by hand; regenerate. Every arm is 140 `valid_seen` games unless stated, paired by ALFWorld game file, tested with an exact McNemar. Confidence intervals are percentile intervals from 10,000 paired bootstrap resamples over game files, seed 20260815.

Baseline replicates: is the control stable?

August 2026, fixed harness. Absolute rates only; no reference arm.

arm	success rate	games
canonical (May 2026, retired)	33.6%	140
replicate 1	39.3%	140
replicate 2	39.3%	140
replicate 3	39.3%	140
replicate 4	41.4%	140

Same-epoch mean 39.8%, spread 2.1pp over 4 replicates.

ALFWorld-trained curator, full fine-tune seed-2, 8B executor

August 2026, contemporaneous control

Reference: 39.3% (`no_memory_8b.jsonl`, n=140 paired games). MDE80 = smallest effect detectable at 80% power.

arm	SR	delta	95% CI	p	p (Holm)	q (BH)	MDE80
ckpt5	32.1%	-7.1	[-13.6, -0.7]	0.0525	0.627	0.315	9.4
ckpt10	37.1%	-2.1	[-10.7, +5.7]	0.7359	1.000	0.928	11.8
ckpt15	36.4%	-2.9	[-10.0, +3.6]	0.5413	1.000	0.928	9.8
ckpt20	40.0%	+0.7	[-7.9, +9.3]	1.0000	1.000	1.000	11.8
ckpt25	32.1%	-7.1	[-14.3, +0.0]	0.0872	0.872	0.349	10.6
ckpt30	31.4%	-7.9	[-15.0, -0.7]	0.0522	0.627	0.315	10.4
ckpt35	42.9%	+3.6	[-4.3, +11.4]	0.4731	1.000	0.928	11.1
ckpt40	41.4%	+2.1	[-6.4, +10.7]	0.7428	1.000	0.928	12.2
ckpt45	37.9%	-1.4	[-8.6, +5.7]	0.8450	1.000	0.928	10.2
ckpt50	35.0%	-4.3	[-11.4, +2.9]	0.3269	1.000	0.928	10.2
ckpt55	35.0%	-4.3	[-12.1, +3.6]	0.3915	1.000	0.928	11.7
ckpt60	37.9%	-1.4	[-9.3, +5.7]	0.8506	1.000	0.928	10.6

* survives Holm correction within this family.

ALFWorld-trained curator, full fine-tune seed-1, 8B executor

August 2026, contemporaneous control

Reference: 39.3% (no_memory_8b.jsonl, n=140 paired games). MDE80 = smallest effect detectable at 80% power.

arm	SR	delta	95% CI	p	p (Holm)	q (BH)	MDE80
ckpt5	30.7%	-8.6	[-15.7, -1.4]	0.0357	0.428	0.214	10.6
ckpt10	43.6%	+4.3	[-4.3, +12.9]	0.4050	1.000	0.540	12.0
ckpt15	34.3%	-5.0	[-12.9, +2.9]	0.2810	1.000	0.540	11.1
ckpt20	34.3%	-5.0	[-12.9, +2.9]	0.2810	1.000	0.540	11.1
ckpt25	35.0%	-4.3	[-12.1, +3.6]	0.3915	1.000	0.540	11.7
ckpt30	30.7%	-8.6	[-15.7, -1.4]	0.0357	0.428	0.214	10.6
ckpt35	37.1%	-2.1	[-9.3, +5.7]	0.7111	1.000	0.776	10.8
ckpt40	45.0%	+5.7	[-2.9, +14.3]	0.2430	1.000	0.540	12.0
ckpt45	40.0%	+0.7	[-7.1, +8.6]	1.0000	1.000	1.000	10.8
ckpt50	33.6%	-5.7	[-12.1, +0.7]	0.1338	1.000	0.535	9.4
ckpt55	42.9%	+3.6	[-4.3, +11.4]	0.4731	1.000	0.568	11.1
ckpt60	35.7%	-3.6	[-10.0, +2.9]	0.4049	1.000	0.540	9.6

* survives Holm correction within this family.

Reasoning-trained curator on ALFWorld, valid-seen split

August 2026, contemporaneous control

Reference: 39.3% (no_memory_8b.jsonl, n=140 paired games). MDE80 = smallest effect detectable at 80% power.

arm	SR	delta	95% CI	p	p (Holm)	q (BH)	MDE80
ckpt5 *	27.1%	-12.1	[-18.6, -6.4]	0.0002	0.002	0.002	9.2
ckpt10	37.1%	-2.1	[-10.0, +5.7]	0.7201	1.000	0.910	11.1
ckpt15 *	26.4%	-12.9	[-20.0, -5.7]	0.0005	0.005	0.003	10.2
ckpt20	37.1%	-2.1	[-10.0, +5.7]	0.7283	1.000	0.910	11.5
ckpt30	35.7%	-3.6	[-10.7, +3.6]	0.4421	1.000	0.872	10.4
ckpt35	40.0%	+0.7	[-6.4, +7.9]	1.0000	1.000	1.000	10.4
ckpt45	38.6%	-0.7	[-8.6, +7.9]	1.0000	1.000	1.000	11.8
ckpt50	43.6%	+4.3	[-2.9, +11.4]	0.3075	1.000	0.769	9.8
ckpt55	36.4%	-2.9	[-9.3, +3.6]	0.5235	1.000	0.872	9.4
ckpt60	33.6%	-5.7	[-12.1, +0.7]	0.1338	1.000	0.446	9.4

* survives Holm correction within this family.

Pending, not yet measured: ckpt25, ckpt40.

Held-out valid-unseen split: curator comparison and content controls

August 2026, contemporaneous control. 134 games.

Reference: 35.1% (`no_memory.jsonl`, n=134 paired games). MDE80 = smallest effect detectable at 80% power.

arm	SR	delta	95% CI	p	p (Holm)	q (BH)	MDE80
Gemini 2.5 Pro curator	41.0%	+6.0	[-3.0, +14.9]	0.2800	0.280	0.280	13.5
trained curator ckpt50	44.0%	+9.0	[+0.0, +17.9]	0.0730	0.146	0.146	12.9

* survives Holm correction within this family.

Pending, not yet measured: trained curator ckpt45, trained curator ckpt55, trained curator ckpt60, ckpt50, shuffled retrieval, hand-written skills (oracle).

32B executor transfer, full fine-tune seed-2 curator

August 2026, contemporaneous control

Reference: 43.6% (`no_memory_32b.jsonl`, n=140 paired games). MDE80 = smallest effect detectable at 80% power.

arm	SR	delta	95% CI	p	p (Holm)	q (BH)	MDE80
ckpt5	54.3%	+10.7	[+2.1, +19.3]	0.0275	0.083	0.041	12.8
ckpt10 *	57.9%	+14.3	[+5.7, +22.9]	0.0017	0.010	0.010	12.3
ckpt15	45.0%	+1.4	[-7.1, +10.0]	0.8679	0.868	0.868	12.0
ckpt20	56.4%	+12.9	[+3.6, +22.1]	0.0114	0.057	0.027	13.6
ckpt25	48.1%	+5.2	[-3.0, +13.3]	0.2962	0.592	0.355	11.9
ckpt30	54.1%	+11.1	[+3.0, +19.3]	0.0135	0.057	0.027	11.9

* survives Holm correction within this family.

Pending, not yet measured: ckpt35, ckpt40, ckpt45, ckpt50, ckpt55, ckpt60.

verl/GiGPO curator, 8B executor

August 2026, contemporaneous control

Not yet measured.